CONCENTRATIONS OF SOLUTIONS 3

Many chemical reactions occur in solutions. There are two parts to a solution: a **solute** and a **solvent**. The solute is the substance being dissolved. The solvent is the substance in which the solute is being dissolved. Water is a very common solvent. Concentrations of solutions can be measured in two ways:

- mass/mass
- mass/volume

Mass/mass states how much of a solute is dissolved in a given mass of solution. A solution of magnesium chloride ($MgCl_2$) can be described as 5 g/100 g. This means that 5 g of the solute (magnesium chloride) has been dissolved in 100 g of solution.

Mass/volume states how much of a solute is dissolved in a given volume of a solution. To make comparing solutions easier, the volume used is usually one litre. A solution of magnesium chloride stated to be 50 g L^{-1} has 50 grams of solute dissolved in each litre of solution.

1 Express each concentration in grams per litre (g L^{-1}).

- **a** 50 g in 100 mL
- **b** 5 g in 500 mL
- **c** 25 g in 250 mL
- **d** 75 g in 2000 mL

Sometimes parts per million (ppm) are used to describe concentrations of solutions. For example, a salt water solution quoted as being 5 ppm salt would be 5 ÷ 1 000 000 or 0.000 5% salt.

2 The active ingredient in a medication is quoted as 120 ppm. How many grams of this substance are in a 50 g tube of the medication? Show your working.

The mole concept is often used in solutions in terms of the **molarity** of a solution. Molarity means the number of moles of solute dissolved per litre of a solution and is sometimes abbreviated to **M**. A solution containing 1 mole of solute per litre of solution is called a 1.0 molar solution or 1.0 M. It can also be shown as 1.0 mol. L^{-1}.

A **mole** of any compound is its **formula mass** expressed in grams.

The molarity of a solution can be calculated using the equation: $\mathbf{C = n \div V}$ where C = concentration (moles per litre), n = number of moles and V = volume (number of litres).

EXAMPLE

i **What is the molarity of a solution 42.40 g of sodium carbonate in 500 mL of water?**

Sodium carbonate = Na_2CO_3

Formula mass = $(2 \times 22.99) + (1 \times 12.01) + (3 \times 16.00) = 45.98 + 12.01 + 48.00 = 106.00$ amu
Hence 1 mole = 106.00 g

Actual mass = 42.40 g. Hence number of moles = 42.04 ÷ 106.00 = 0.40 moles

Express in moles per litre: 0.40 moles is dissolved in 0.50 litre (500 mL); hence the solution is: 0.40 ÷ 0.50 = 0.80 molar

3 Calculate the mass of sodium chloride (NaCl) dissolved in 1 litre of a 5 molar solution.

4 Calculate the mass of sodium hydroxide (NaOH) dissolved in 300 mL of a 2 molar solution.

5 Calculate the molarity of each solution.

- **a** 9.52 g of magnesium chloride ($MgCl_2$) dissolved in 1 litre of solution.
- **b** 9.8 g of sulfuric acid (H_2SO_4) dissolved in 500 mL of solution.
- **c** 21.2 g of sodium carbonate (Na_2CO_3) dissolved in 2 litres of solution.

6 Calculate the mass of sulfuric acid in 250 mL for each given concentration.

- **a** 1.0 molar solution
- **b** 0.5 molar solution
- **c** 3.0 molar solution

7 Explain the concentration descriptions *mass/mass* and *mass/volume*.

8 Define *molarity*.

ISBN 978 1 92561 066 6
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